Polytechnic University of the Philippines

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College of Computer and Information Sciences

Foodietection: Classifying Go, Grow, Glow Foods for Filipino Nutrition Application using MobileNetV3 and Fuzzy Logic Food Recommender for Food Information Dissemination

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Chapter 1

THE PROBLEM AND ITS SETTING

1.1 Introduction

A very important key to managing and keeping your health in good status is the effective management of food nutrition in our food intake. Our food choices and the nutrition are crucial in preventing diseases. Knowing the right food to eat can help you be able to gain strength to cope with illnesses you might have. A good way of understanding nutrition can provide you with information on what can help you maintain good health or even improve your state. Nutrition in foods is the way to get the energy for our body to function. We require constant supply of nutrients in our bodies every day (Case-Lo, 2017).

One of the main problems is that information about Food Nutrition is easily searchable throughout the internet, yet there seems to be no effect or visible effect on how people balance their food nutrition throughout the day as seen on the key reports of Worldbank.org. In the Philippines, parents (usually those with children) are the ones who celebrate Nutrition Month every July in Schools. According to Philippines National Nutrition Council, celebration of Nutrition Month is a way to raise awareness to families about strategies on having a proper and healthy diet to prevent malnutrition. Part of the need to celebrate Nutrition Month every July is because a lot of children usually do not have self-control to eat healthy and just eat whatever they liked. In addition to that, there is also a dire need to spread information about food nutrition especially in places where balanced diet and food diversity is scarce so as to give them new ideas on how to assemble their balanced diet through different food categories such as Go, Grow, and Glow foods.

Each food group serves a specific purpose in giving nutrients that keep the body healthy. Carbohydrate-rich meals is referred regarded as "go food." Protein-rich foods

make up Grow food. Glow foods include vegetables and fruits, which are high in vitamins and minerals. Even though Philippines has plenty of this kinds of foods, previous studies have shown that the dietary diversity of young Filipinos is limited, and that the prevalence of nutrient inadequacies is high according to (Mak, Angeles-Agdeppa, Lenighan, Capanzana, & Montoliu, 2019). Their study shows that the lack of dietary diversity is one of the reasons for the low probability of adequate intakes of micronutrients, particularly those from poorer households. A study conducted showed that the awareness and adherence to food based dietary guidelines among household meal planner in the Philippines is low and there is a need to strengthen and harmonize the dissemination efforts of food based dietary guidelines not only to meal planner but also among various segments of the population using various media channels (Lopez-Madrid et al., 2018). Therefore, researcher aims to produce another media for helping meal planner to prepare better meal.

Nowadays, people like to take pictures of their food as a form of ritual before eating. It is a form of a ritual behavior that apparently “adds more enjoyment and taste” to their food. According to Vohs et al. (2013) through their experiment 1: *Do Rituals Enhance Consumption?* supported their hypothesis that rituals do indeed increase consumption. They demonstrated the result of the participants performing rituals like taking pictures of their self-enjoyment that results in their behavior of savoring the food. Food photos are very popular because they are both easy to do and is relevant to everyone. Therefore, we would like to exploit this popular culture of taking pictures of food to further disseminate information about the daily food nutritional guide as a solution for the inadequacy of nutrients taken by Filipinos due to lack of information regarding the daily food nutritional guide.

According to SightCorp (n.d), “Image recognition is the ability of a computer powered camera to identify and detect objects or features in a digital image or video. It is a method for capturing, processing, examining, and sympathizing images”. Image recognition 3 general steps namely, pre-processing, feature extraction, and learning algorithm for classification (Mallick, 2016). Pre-processing is the first step in image recognition that formats images before they are used by learning algorithms and this includes resizing, orienting, and correcting the color of an image (Nelson, 2020). The second step in image recognition is feature extraction and according to MathWorks (n.d) “Feature extraction is process of transforming raw data into numerical features that can be processed while preserving the information in the original data set”. Finally, a machine learning algorithm is used in order train a model to recognize images.

According to MathWorks (n.d), convolutional neural network (CNN or ConvNet), is a network architecture for deep learning which learns directly from data, eliminating the need for manual feature extraction. CNNs are particularly useful for finding patterns in images to recognize objects, faces, and scenes. They can also be quite effective for classifying non-image data such as audio, time series, and signal data. MobileNetV3 is a convolutional neural network that is tuned to mobile phone CPUs through a combination of hardware-aware network architecture search (NAS) complemented by the NetAdapt algorithm, and then subsequently improved through novel architecture advances.

Fuzzy logic is a computing approach based on "degrees of truth" rather than the standard "true or false" (1 or 0) Boolean logic used by modern computers. In the 1960s, Lotfi Zadeh of the University of California in Berkeley proposed the concept of fuzzy logic according to (Chai, 2019). Fuzzy logic mimics human behavior for reasoning and decision-making and treat ambiguous phenomena mathematically and express the degree of

incomprehensibility in human thinking along with connecting it to a real number according to (Vivek, Subbarao, Routray, Kamini, & Dash, 2019).

In relation to the study discussed, the proponents of the study thought of combining the art of photo-taking and food detection using computers. The aim is to be able to capture a photo and detecting food inside the photo and output information about the food that was captured, showing relevant information about the food in the image such as nutritional values. The study proposed aims to design a mobile-based system that classifies food into the Go, Grow, and Glow using MobileNetV3 and Fuzzy Logic and recommend foods that can complete their suggested daily nutritional intake. This will help people who has their target nutrient values to consume as the system will output nutrient values and give the experience of food-taking rituals from the camera function. The proponents also hope that this study will provide future reference materials to future research and improve classification on Go, Grow, Glow foods through other methods other than Fuzzy Logic.

1.2 Theoretical Framework

Artificial Neural Networks. This paper is largely based on artificial neural networks (ANN). In theory, ANNs can stimulate the asynchronous activity of the human nervous system (Meyers, 2001). Artificial Neural Networks are technologies derived from the brain and the nervous system. These networks mimic biological neural systems but in a reduced set of constructs derived from biological neural systems. ANN models can simulate electrical activity of the brain and the nervous system such as when our eyes identify certain objects. ANNs

Image Segmentation. According to Ghosh et al., (2020), “Image segmentation can be defined as a specific image processing technique that is used to divide an image into two or more meaningful regions. Image segmentation can also be seen as a process

of defining boundaries between separate semantic entities in an image. From a more technical perspective, image segmentation is a process of assigning a label to each pixel in the image such that pixels with the same label are connected with respect to some visual or semantic property". In this paper, semantic segmentation will be utilized to differentiate regions in an image. Particularly the LR-ASPP (Lite R-ASPP) as defined in (Howard et al., 2019). This method utilizes R-ASPP – a reduced design of the Atrous Spatial Pyramid Pooling module (Chen et al., 2018), that has two branches consisting of 1x1 convolution and a global-average pooling operation (Liu et al., 2015). LR-ASPP deploys the global-average pooling in which a large pooling kernel with a large stride (to save some computation) and a one 1x1 convolution is utilized. In addition to this, an atrous convolution is applied to the last block of the mobilenetV3.

Noise Reduction. According to Ali & Verma, (2013), noise removal is the process of removing or reducing the noise from the image. The noise removal techniques reduce or remove the visibility of noise by smoothing the entire image leaving areas near contrast boundaries. To reduce the noises present in the datasets that will be used, the researchers will implement the median filter- a non-linear and order-statistic method. The principle of the median filter is replacing the gray level of each pixel by the median of the gray levels in a neighborhood of pixels, instead of utilizing the average operation (Tan & Jiang, 2019).

Feature Extraction. According to DeepAI (n.d), feature extraction is a process of dimensionality reduction by which an initial set of raw data is reduced to more manageable groups for processing. A characteristic of these large data sets is many variables that require a lot of computing resources to process. Feature extraction is the name for methods that select and /or combine variables into features, effectively reducing the amount of data that must be processed, while still accurately and completely describing the original data set.

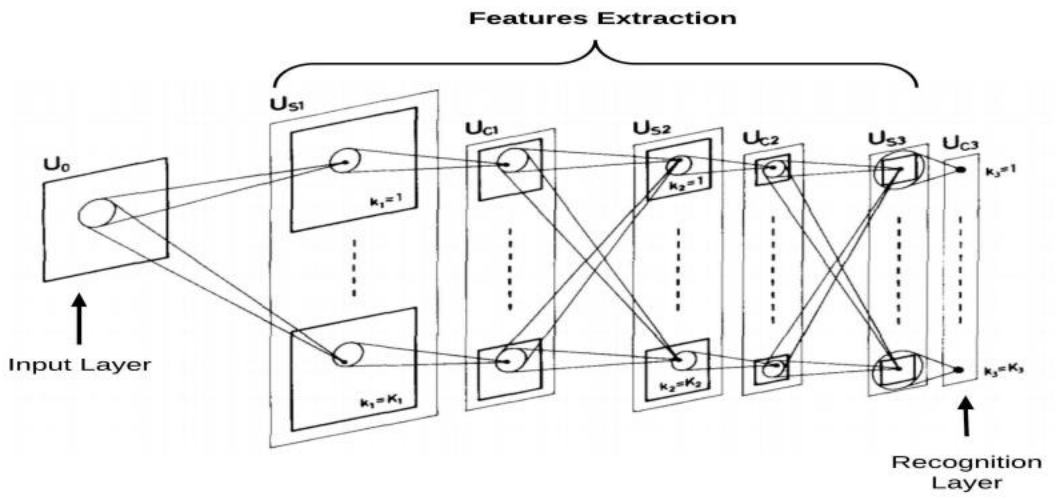

Figure 1. Feature Extraction Process

Deep Learning. According to (LeCun, Bengio, & Hinton, 2015), deep learning allows computational models made up of many processing layers to learn multiple levels of abstraction for data representations. Deep learning uses the backpropagation algorithm to show how a machine should adjust its internal parameters that are used to generate the representation in each layer from the previous layer. Deep convolutional nets have revolutionized image, video, voice, and audio processing, whilst recurrent nets have shed light on sequential data like text and speech.

Convolutional Neural Network. According to Yamashita et al. (2018), “Convolutional Neural Network is a type of deep learning model for processing data that has a grid pattern, such as images, which is inspired by the organization of animal visual cortex and designed to automatically and adaptively learn spatial hierarchies of features, from low to high-level patterns. Convolutional Neural Network is a mathematical construct that is typically composed of three types of layers (or building blocks): convolution, pooling, and fully connected layers. The first two, convolution and pooling layers, perform feature extraction, whereas the third, a fully connected layer, maps the extracted features into final output, such as classification.”

Convolutional Neural Network algorithm has two main processes: convolution and sampling. Convolution process: use a trainable filter Fx, deconvolution of the input image (the first stage is the input image, the input of the after convolution is the feature image of each layer, namely Feature Map), then add a bias bx, we can get convolution layer Cx. A sampling process: n pixels of each neighborhood through pooling steps, become a pixel, and then by scalar weighting Wx + 1 weighted, add bias bx + 1, and then by an activation function, produce a narrow n times feature map Sx + 1.

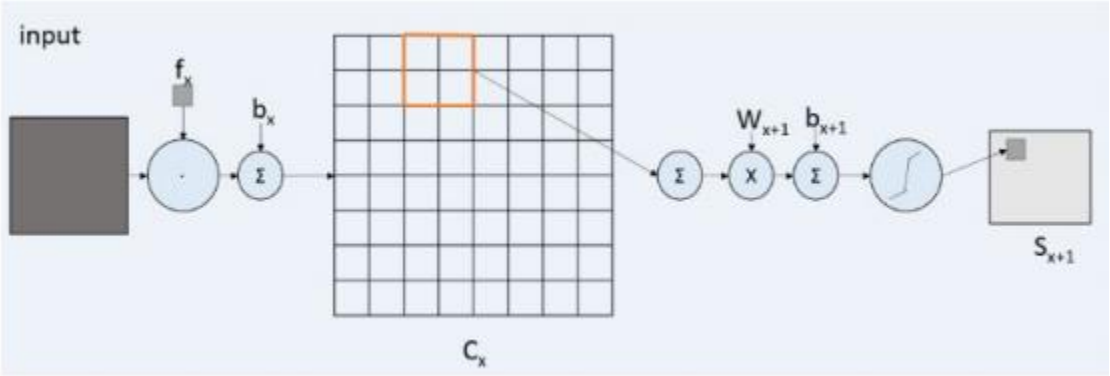

Figure 2. Sampling Process

CNN Architecture. CNN design algorithm need experience in architecture design, and need to debug unceasingly in the practical application, to obtain the most suitable for a particular application architecture of CNN. Based on gray image as the input of 96 96, in the pre-process stage, turning it into 32 32 of the size of the image. Design depth of the layer 7 convolution model: input layer, convolution layer C1, sub sampling layer S1, convolution layer C2, sampling layer S2, hidden layer H and output layer F.

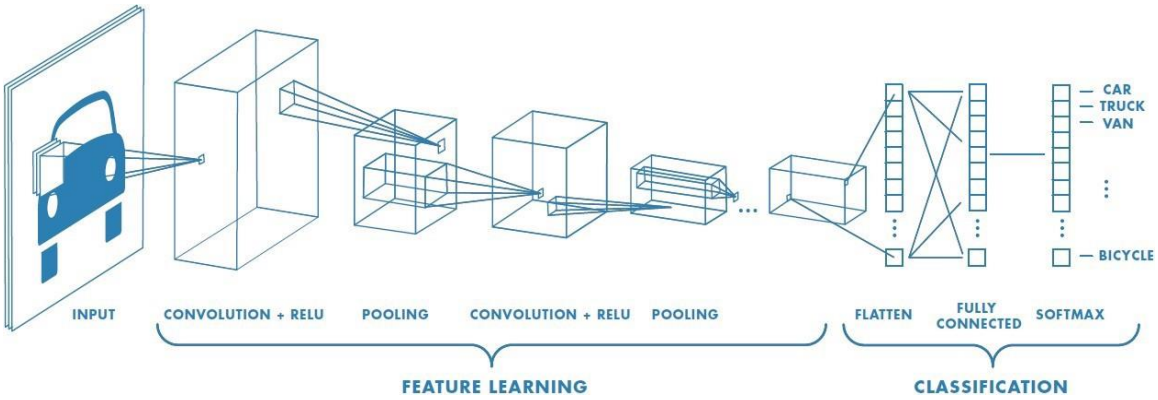

Figure 3. CNN Architecture

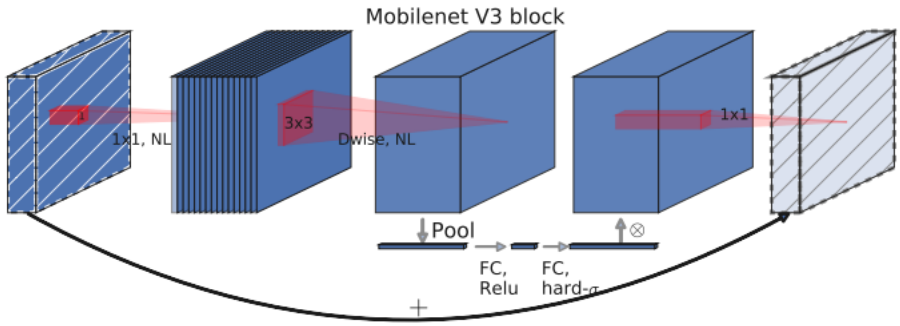

Figure 4. MobileNetV3

MobileNetV3. MobileNetV3 utilizes Inverted Residual and Linear Bottleneck designed layers along with a Squeeze and Excite approach in residual layers to reduce the parameters of the network. This model also uses nonlinearities on certain layers of the network. MobileNetV3 also utilizes platform-aware NAS and NetAdapt algorithm to optimize the model based on a given hardware platform.

Fuzzy Logic. Fuzzy logic is an extension of Boolean logic which is based on the mathematical theory of fuzzy sets, which is a generalization of the classical set theory. By introducing the notion of degree in the verification of a condition, thus enabling a condition to be in a state other than true or false, fuzzy logic provides a very valuable flexibility for

reasoning, which makes it possible to consider inaccuracies and uncertainties (Dernouncourt, 2013).

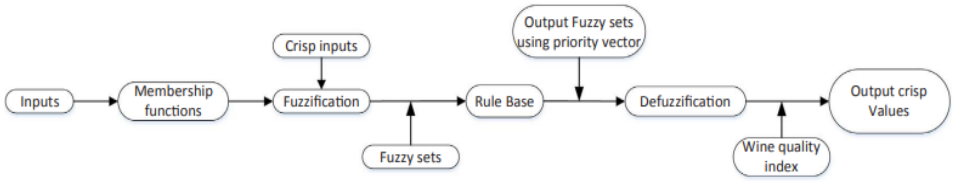

Figure 5. Wine Quality Classification using Fuzzy Logic

In the study of Petropoulos et al., (2017), they created a fuzzy logic multi criteria decision making system. The figure above shows the model created for their study. The researchers will use fuzzy logic in order to classify whether a food belongs to go, grow, or glow based on its nutritional content.

1.3 Conceptual Framework

The proponents of this study have noticed the ever-growing trend of image recognition in the field of Machine Learning and Data Science about Food. However, although there are number of studies related to food recognition and nutrition-focused research, there are still very limited resource when it comes to looking for Go, Grow, Glow foods in the Philippines. That is why *Foodietection: Classifying Go, Grow, Glow Foods for Filipino Nutrition Application using MobileNetV3 and Fuzzy Logic Food Recommender for Food Information Dissemination* was developed to add more variety to currently existing studies and inspire more studies to go in the direction for improving Food Identifier and Recommendation for children.

This part will show the concept of the proponent's proposed system for Classifying Go, Grow, Glow using MobileNetV3 and Fuzzy Logic

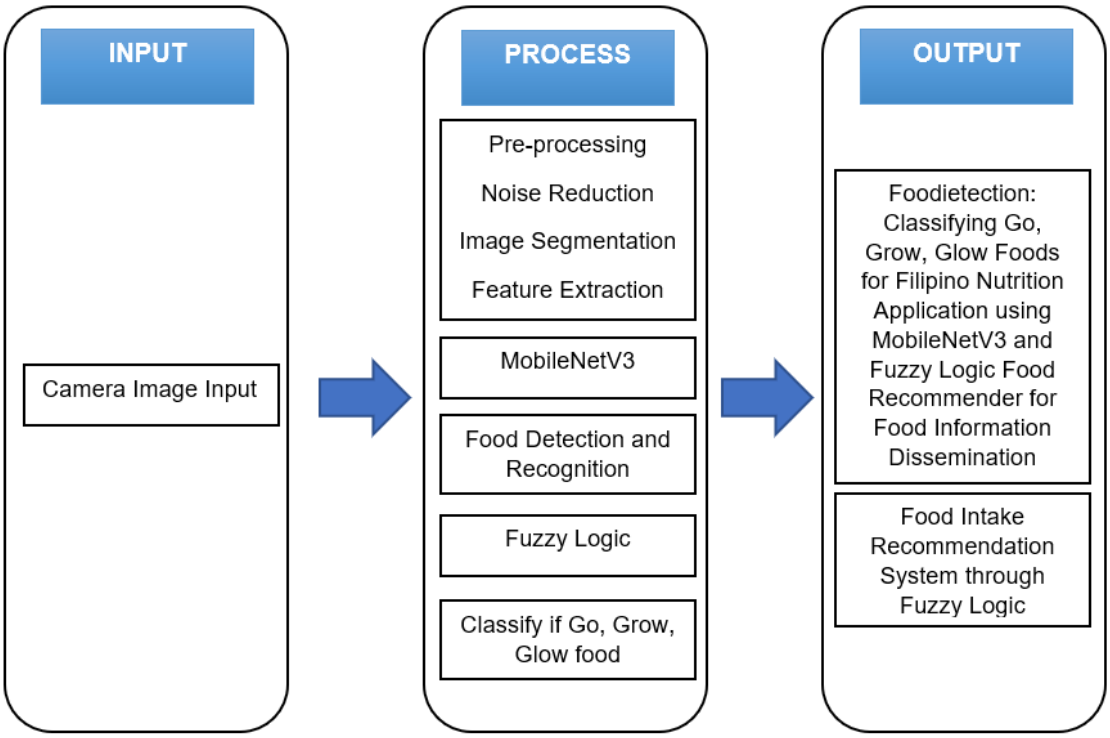

Figure 6. Conceptual Framework of the Study

Input – The input of the study will only include Camera Input Image since the proposed application will require the use of camera for succeeding processes to occur.

Process – The processes of the study will include multiple techniques to process the input image for detection and classification. The processes include Pre-processing techniques including Image Segmentation and Feature Extraction, MobileNetV3 for Image Detection and Fuzzy Logic for Classifying if the detected food is either Go, Grow, or Glow food.

Output – After going through the input and process of the system, the corresponding output would be the Mobile application of *Foodietection: Classifying Go, Grow, Glow Foods for Filipino Nutrition Application using MobileNetV3 and Fuzzy Logic Food Recommender for Food Information Dissemination*

1.4 Statement of the Problem

This part presents the proponent's statement of the problem and the goal of the researchers to be determined after doing the Thesis. This study aims to solve the problem of not having a way to classify Go, Grow, and Glow foods. Go, Grow, Glow are classified through their nutritional content rather than their physical appearance hence CNN cannot classify if a food belongs to Go, Grow, and Glow food category. Most people only know that a specific food belongs to their specific category, but not what exactly what the food contains to qualify for a specific food category. This problem will be solved by using MobileNetV3 to detect and recognize the specific food and using Fuzzy Logic to classify if the recognized food belongs to either Go, Grow, or Glow food. Specifically, the study seeks to answer the following problems:

1) What is the performance evaluation of the system in classifying food groups in terms of:

   1.1) Precision

   1.2) Recall

   1.3) F-1 Score?

2) What is the accuracy of the system in identifying the different foods using the MobileNetV3 Model?
3) What is the accuracy of the system in classifying the different food groups using Fuzzy Logic?
4) What is the accuracy of the system in recommending food intake for the users?
5) Is there a significant difference in detecting food image using MobileNetV2 and when using MobileNetV3?

1.5 Hypothesis

Null Hypothesis

H_0: There is no significant difference in detecting food image using MobileNetV2 and when using MobileNetV3.

1.6 Scope and Limitations of the Study

This part will present the scope of which the proponents will present on the development of this Thesis. This part will also include the limitations of the study and restrictions that the proponents have faced.

Foodietection is a mobile-based application system that classifies food specifically on go, grow, and glow. Food Nutrition Guides and Food Datasets would prioritize Filipino Data as it is the most accessible to the proponents. This study will also focus on 39 different food types as shown in appendix C.

The study focuses on the use of MobileNetV3 and Fuzzy Logic in categorizing the food into their respective food groups. It will also focus on the recommendation of the missing food group in the meal being captured.

The researchers will develop a mobile application for android devices only. The said android device should also have a working camera. The proposed system will capture the food using the device's camera. The food should have enough lighting for the system to work. The camera angle should also be in high view.

The system is limited in only showing the nutritional content of the food detected and where the food belongs in the food group chosen by the researchers (Go, Grow, and Glow).

1.7 Significance of the Study

In this part, the proponents presented the significance and why this study is important to other social aspects of the community.

Researchers, they may be able to use relevant information presented in this study to further improve their research output and offer a much-improved study on the subject matter. This study will utilize fuzzy logic in recommendations for the food intake of its users, thus, this will help other researchers to find knowledge for fuzzy logic as a method in recommendation systems. This also utilizes the latest version of mobilenetv3 to which there is a limited study as of now. This will further increase the current knowledge in the accuracy of the mobilenetv3 in application to food identification.

Healthcare Workers, especially nutritionist and dietitians, to help them have an easier time in explaining food nutrition contents and intake recommendation to their patients.

Students, who might be able to get inspiration from this study for their own additional knowledge. Students can also use this study as a source of knowledge to food detection using convolutional networks and fuzzy logic in food intake recommendations.

Society, everyone can be the beneficiaries of this study to help them give sufficient knowledge about Go, Grow, and Glow foods and for a much easier understanding of the daily food nutrition guide of the FNRI. Giving meal planners and consumers information about the food to prepare and food to eat to ensure a nutritious meal. They are reason why the proponents have developed this study about Foodietection.

Especially, this study can benefit **Children** and **Teenagers** since they are at the age where food nutrition is very important and is being taught at their respective schools every month of July as part of the Nutrition Month where the theme is about Go, Grow, and Glow. This does not limit the information benefit that it will give to children and teenagers but also contribute to food nutrition awareness to the general public including the adults.

1.8 Definition of Terms

To further elaborate on the study, the proponents have compiled a list of terms that was used as a technical term for this study.

Confusion Matrix – a technique for summarizing the performance of classification algorithms/methods. It contains *TP*, *FP*, *FN*, *TN*.

Convolutional Neural Network (CNN) – one of the more common and known deep learning algorithm that places weighted values and bias for easier differentiation.

Deep Learning – allows computational models made up of many processing layers to learn multiple levels of data abstraction for data representations.

False Negative – If the actual class is true but the prediction is false

False Positive – If the actual class is false but the prediction is true

Feature – a piece of information about the content of an image

Feature Extraction - is a process of dimensionality reduction by which an initial set of raw data is reduced to more manageable groups for processing

Fuzzy Logic – used to categorize the food into either Go, Grow, or Glow

Glow foods – Glow foods contain the vegetables and fruits and these help the body develop immune system through vitamin and mineral supply to keep the body healthy and properly functioning.

Go foods – Go foods are the type of food that provide fuel and help people ‘go’ to be active. These foods give our muscles fuel to run, jump, swim, cycle, and our brain fuel to concentrate.

Grow foods – Grow foods help the body grow bigger and stronger hence the word ‘grow’. They build our body’s bone, teeth, and muscles. They help people feel full and not get hungry easily.

Image Segmentation – for dividing an image to get its detailed features | Semantic segmentation describes the process of associating each pixel of an image with a class label.

NetAdapt – this approach is a complementary approach for platform-aware NAS that fine-tunes individual layers in a sequential manner instead of a crude but global architecture.

MobileNetV3 - is a convolutional neural network that is tuned to mobile phone CPUs

Picture-culture – the taking of photos of foods or drinks before actually consuming it.

Platform-aware NAS – a platform-aware neural architecture approach in finding global network structures

True Negative – If the actual class is false and the prediction is false

True Positive – If the actual class is true and the prediction is true

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

In this chapter, the proponents will present relevant information and studies that they have gathered throughout the development of this study. This chapter will present past studies that will help the reader gain more information about the flow of this study.

2.1 Deep Learning

According to (LeCun, Bengio, & Hinton, 2015), deep learning allows computational models made up of many processing layers to learn multiple levels of abstraction for data representations. These techniques have significantly enhanced the state-of-the-art in speech recognition, visual object recognition, object detection, and a variety of other fields, including drug development and genomics. Deep learning uses the backpropagation algorithm to show how a machine should adjust its internal parameters that are used to generate the representation in each layer from the previous layer. Deep convolutional nets have revolutionized image, video, voice, and audio processing, whilst recurrent nets have shed light on sequential data like text and speech. Nowadays, deep learning can be found in different fields. In medical field, intelligent imaging and deep learning used for analysis and early diagnosis of disease (Cai, Gao, & Zhao, 2020). One specific class of deep learning is Convolutional Neural network (CNN) (Kamilaris & Prenafeta-Boldu, 2018).

2.2 Fundamentals of CNN

Convolutional Neural Network or CNN is one of the more common and famous Deep Learning Algorithm that takes an image as the input, then places weighted values and bias to different aspects/objects in the image to help differentiate images from each other. According to Hussain et al. (2019), *Convolutional*, *activation*, *pooling*, *fully connected (FC)*, and softmax layers are the five (5) primary layers in CNN models. The convolutional layer has a set of filters that each activating certain features from the image.

The filters of each convolutional layer hold the features of input image such as shapes, blobs, edges, etc. The second layer, the activation layer, also known as *Rectified Linear Unit* (ReLU) activates the neuron after the computation of a nonlinear function of the given input. The third layer, pooling layer whittles down the number of parameters by reducing the spatial size of the network or the input. The fourth layer − *fully connected* (FC) layer is basically the hidden layers of that are common in neural networks. It shows the aggregation of the information/features from all the convolutional layers that is scanned before it. The fifth layer, softmax layer, is the normalization and prediction that enables generation of output as probabilities.

2.3 Applications of CNN

CNN is a machine learning approach that has been around for long. Its first ever use can be traced back to the late 90s (Lecun, Bottou, Bengio, & Haffner, 1998). From then, CNN has been slowly evolving with its techniques and approaches. Slowly and gradually CNNs success starts becoming profound.

2.4 Recent breakthrough in CNN

Recent works showed evidence that network depth is a crucial factor for Convolutional layers. By increasing the depth of weight layers (Simonyan & Zisserman, 2015), classification accuracy also increases. It also appeared to the neural network *Inception* (Szegedy, et al., 2015). Inception was designed to handle increasing network depth and width while keeping the computational cost constant.

These evidences brought the question: *Does more stack of layers means a better learning network?* Apparently, when deep networks start converging, a *degradation* problem arises. As the network depth increases, the accuracy gets saturated and rapidly degrades. This degradation is not due to overfitting (He & Sun, Convolutional Neural Networks at Constrained Time Cost, 2015). Additionally, Srivastava et al. (2015) reported

that adding additional layers to a deep model leads to higher training error. Due to these concurrent issues brought by deeper neural networks, Kaiming et al. (2016) presented a new architecture, which produced an outstanding result − the introduction of Residual Networks (ResNet). ResNet tackled the issue underlying the network depth of CNNs. Through ResNet, deeper networks can now perform accurately.

2.5 Applications of CNN in Food Detection

CNN is one of the best deep learning algorithms to use in image detection (Leonard, 2019). It outperforms other methods such as manual feature extractors, and conventional machine learning algorithms (Zhou et al., 2019).

In the study done by Park et al., (2019), they created a mobile based food image detection and recognition model of Korean foods for dietary management. The proposed DCNN implemented a wider and deeper network to improve performance. The results showed that the proposed DCNN model was more accurate in determining different Korean foods compared to other state-of-the-art models (AlexNet, GoogleNet, ResNet, etc). The proposed DCNN model had a test accuracy of 91.3% and a prediction time of 0.42ms Although the proposed DCNN model was more accurate, its loss function was too noisy and is not capable of smooth reduction due to the use of data augmentation that produced very similar images in their training data sets. They suggested to use different sources of images from the data set to better improve the model.

The study by Shen et al., (2020) aims to create a mobile based application that will help identify the food the user photographs and its nutritional content. The proposed system uses CNN as its machine learning algorithm to recognize the food. The system also created a web crawler that will help determine what the nutritional content of the food was photographed. Test results show that the system had an accuracy of 85% but failed to identify and process mixed physical images.

A study conducted by (Subhi & Ali, 2018) for food detection and recognition, Deep Convolutional Neural Networks are used. They employ a new data set culled from the Internet of the most popular local Malaysian foods. The results reveal that CNN is significantly more accurate than a traditional approach based on manually derived features. It also demonstrates that the feature map is dominated by food colors. A study done by (Attokaren, Fernandes, Sriram, & Koolagudi, 2017) to categorize food from input photos using Convolutional Neural Networks. The study's findings indicate that CNNs are acceptable in terms of usage, but that they require a powerful computer to experiment on large multi-media datasets. The training of a CNN takes more computing time, however once the system is correctly trained, performance is critical.

Interestingly, most of the food detection systems present, use on-the-shelf approaches. In the study of Hui et al. (2016), they incorporated semantics in food detection as their approach. They treated the food recognition problem as a *fine-grained problem*. Their study resulted to an increase in the accuracy of food detection but is limited to the number of food type included in their semantic analyzer.

Using a *slice convolution block* partnered with a residual network, Niki et al. (2018), along with the other different approaches in food recognition, achieved the highest accuracy. Their system which they called WISeR achieved a Top-1% of 90.27. Through adding the information of the *vertical layer* of foods, via using the slice convolution block, they could come up with a more accurate classification of foods. Although their system has the highest accuracy, due to its intricate processes, the WISeR takes up lot of computational power and is hard to implement on devices such as smart phones or tablets wherein the system could have been used effectively.

In terms of Feature Extraction, in a study done by Liu, Pu, and Sun (2021), they stated that Feature Extraction is a very vital step in image processing. Also, it is important

to note that the usual CNN models go for 2-Dimensional convolutional filters to analyze images that are in grayscale and RGB (red, green, blue) images. This is not to say that that there are no other types of CNN models as there are still 1-Dimensional and 3-Dimensional filters. Features extracted from 1-D, 2-D, 3-D CNN are used to train classifiers such as SVM (Support Vector Machine) and K-Nearest Neighbor (KNN). In the study, 1-D CNN filters were used for signal processing like Speech Recognition and Noise Filtering. 2-D CNN filters were used for extracting RGB features in the food image. Although 1-D and 2-D can catch spectral and spatial features respectively, they both cannot get the joint spatial-spectral without using individual 1-D and 2-D CNN. 3-D CNN can utilize these 3-D kernels to capture both spatial-spectral features simultaneously without needing any pre- or post-processing.

According to (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018), neural networks enable superhuman accuracy for challenging image recognition tasks however the drive to improve accuracy often cost a lot. In the last several years, numerous teams has lead dramatic improvement in neural network design such as AlexNet, VGGNet, GoogleNet and ResNet. In the study of (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018), they describe new neural network architecture that is design for mobile called MobileNetV2. It is specifically tailored for mobile and resource constrained environments. Their network pushes the state of the art for mobile tailored computer vision models, by significantly decreasing the number of operations and memory needed while retaining the same accuracy.

MobileNet model become useful to mobile based applications in the past few years. In the study of (Velasco, et al., 2019), they use MobileNet for skin disease classification. The result of their system achieved 94.4% accuracy in determining seven

skin disease. Another study conducted by (Akay, et al., 2021) uses MobileNetV2 Model in determining systematic sclerosis skin and achieved 94.8% accuracy.

2.6 Fuzzy Logic

Fuzzy logic is a computing approach based on "degrees of truth" rather than the standard "true or false" (1 or 0) Boolean logic used by modern computers. In the 1960s, Lotfi Zadeh of the University of California in Berkeley proposed the concept of fuzzy logic according to (Chai, 2019). Fuzzy logic mimics human behavior for reasoning and decision-making and treat ambiguous phenomena mathematically and express the degree of incomprehensibility in human thinking along with connecting it to a real number according to (Vivek, Subbarao, Routray, Kamini, & Dash, 2019). In the study done by (Jain & Gupta, 2018), they explored the use of fuzzy logic in recommender systems. They found that fuzzy logic-based recommender systems help deal with uncertainty of data.

In study done by (Reddy & Khare, 2016), they used fuzzy logic to predict diabetes. The results indicate that the proposed classification framework has outperformed by attaining higher accuracy, specificity, and sensitivity when compared with the existing approaches. Another study by Aamir et al., (2021), used fuzzy rule-based system to create a classification model for the types of diabetes and predict early signs of it. They first created two different fuzzy classifiers then created fuzzy rules based on the classifiers. Results showed that their proposed system scored 96.47% and 95.38% accuracy, respectively. Another study conducted by (Matiko, Beedy, & Tudor, 2014) use fuzzy logic to classify negative and positive emotions. The result of the algorithm achieved 63.52% accuracy that is higher than widely used algorithms.

A study conducted by (Nilashi, Ibrahim, Ahmadi, & Shahmordi, 2017) created fuzzy rules for the classification of objects using Classification and Regression Trees (CART) in the knowledge-based system of fuzzy rule-based reasoning method for breast cancer

sickness. They devised a strategy to deal with the problem of multi-collinearity, PCA should be incorporated into the suggested knowledge-based system. Wisconsin Diagnostics' experimental results Breast cancer and mammographic mass datasets reveal that the proposed strategies improve the accuracy of breast cancer prediction significantly. The proposed knowledge-based system can be utilized to aid medical practitioners in making clinical decisions in the practice of healthcare.

A study done by Taneja et al., (2016), created a classification algorithm based on fuzzy logic. The proposed algorithm was used to classify admission status of students from various universities. The classification was divided into 3 parts, admitted, rejected, and those who are on the fence of being admitted or rejected. The researchers defined the degree of membership for the third classification using ranking and fuzzy logic. *Figure 5* below shows the results of the study. The proposed system of Taneja et al., (2016), scored a much higher accuracy than standard algorithms, Crisp KNN and Fuzzy C-means.

Data Set	Crisp KNN	FCM	Proposed
Admission Data Set	82.59%	85.63%	92.47%
UCI University Data Set	80.32%	83.36%	88.72%

Figure 7. Comparison of Accuracy

In the study done by Murmu and Biswas (2015), they reviewed the application of fuzzy logic together with neural networks to classify crops. The results of their study found that fuzzy logic is an improvement to other supervised learning methods in terms of time and efficiency. Also, they found out that neuro-fuzzy systems perform better in some situations than other classification techniques.

In another study by Petropoulos et al., (2017), they used fuzzy logic to classify wine quality. Although some of the parameters were not included due to unfeasibility, their results show that the created fuzzy system ranked the wines in a similar manner with wine experts. Despite the weaknesses, it is possible to estimate wine quality from easily measurable grape parameters applied in a fuzzy logic system.

In the proposed system by Sangatash et al. (2012), they used Fuzzy logic to classify raw milk based on microbiological and physicochemical qualities, and a fuzzy logic system was developed using a fuzzy inference engine. The input parameters were as follows: total aerobic microorganism count, somatic cell count, coliform count, fat and solid nonfat percentages. The classification input data used to classify the raw milk were according to this scale: Excellent, Very Good, Good, Bad, Very Bad. The system used Mamdani fuzzy, and the results showed that fuzzy logic was successfully applied and served as a decision support technique in determining raw milk grades in terms of quality.

	Fuzzy Logic Prediction							
Expert Knowledge	Grade of Raw Milk Quality	Excellent	Very good	Good	Bad	Very Bad	No. of Raw Milk	Correctly Classified Raw Milk (%)
	Excellent	5	2				7	71.43
	Very Good	1	14	1	1		17	82.35
	Good		2	20	2		24	83.3
	Bad			1	17	2	20	85
	Very Bad				2	10	12	83.3
	Total Observed	6	18	22	22	12	66*/80	
	%	83.3	77.8	90	77.3	83.3		82.5

* Number of raw milk samples correctly classified by proposed fuzzy system

Figure 8. Proposed FIS and expert knowledge in grading raw milk quality

They suggested that combinations of fuzzy logic system with artificial neural networks and genetic algorithms are suggestions to increase accuracy.

Meanwhile, in a paper written by Hmid & Jemaa (2006), they used Fuzzy rules together with image segmentation for Human Face Detection system. Converting the image from RGB to YCbCr eliminates the variance of luminance from the lighting

conditions as much as possible. Then using the Takagi-Sugeno fuzzy inference system (FIS) consisting of two inputs (Cb and Cr) and one output (the decision: skin or non-skin color. The fuzzy rules used are:

1. IF Cb is Light and Cr is Light THEN the pixel =0

2. IF Cb is Light and Cr is Medium THEN the pixel =0

3. IF Cb is Light and Cr is Dark THEN the pixel =0

4. IF Cb is Medium and Cr is Light THEN the pixel =0

5. IF Cb is Medium and Cr is Medium THEN the pixel =1

6. IF Cb is Medium and Cr is Dark THEN the pixel =1

7. IF Cb is Dark and Cr is Light THEN the pixel =0

8. IF Cb is Dark and Cr is Medium THEN the pixel =1

9. IF Cb is Dark and Cr is Dark THEN the pixel =0

They used these rules together with Edge Detection and Ellipse Detection to reduce the number of false alarm detections by 11.59% and a detection rate of 92.08%. The improvement is due to Ellipse detection eliminating many of the skin region that does not have elliptic shape. They concluded that a new algorithm using Fuzzy classification, edge detection and shape analysis has been proposed and the results were satisfactory.

	Faces		Non-Faces	
	Real positives	False positives	Real negatives	False negatives
Level1	79,66	14,00	6	0,33
Level2	73,66	1,66	12,66	6,33

	Detection ratio	False alarms
Level1	99,58	70,00
Level2	92,08	11,59

Figure 9. Performance of Hmid and Jemaa's System

In the study done by Cintra et. al., (2009), they compared different machine learning algorithms to fuzzy logic systems for classification. They used 5 machine learning algorithms for classification, namely: J4.8 Decision Tree, Zero Rules, NaiveBayes, OneRule, and Multilayer Perception. The 2 fuzzy system used are DoC-based Method (which uses Genetic Fuzzy System) and Wang & Mendel Method. Their results show that DoC-based method performs well in terms of accuracy and interpretability. J4.8 performs well in terms of interpretability but having low accuracy. The MLP and NaiveBayes both obtained good accuracy, but their results are not interpretable. The Wang & Mendel method also had good accuracy, but it generates a high number of rules compared to other methods. Finally, both OneRule and Zero Rules both had poor accuracy.

	DoC-Based	WM	Zero Rules	J4.8	NaiveBayes	OneRule	MLP
Position	1	4	7	3	5	6	2
Ranking Average	1.14	4.13	7.00	3.37	4.14	5.00	3.00

Figure 10. Ranking for Classification.

2.7 Convolutional Neural Network and Fuzzy Logic

In the study done by Olaes et. al., (2020), they created bell pepper and chili pepper classifier using convolutional neural network and fuzzy logic. Their results showed that the proposed system was able to fully recognize the difference between bell and chili pepper.

2.8 Food Nutrition Guide

2.8.1 Daily Food Intake Guidelines

Our body needs nutrients to function well. There are more than 40 nutrients that are essential to the body to function well. These nutrients depend on each other, and no single food can provide all these nutrients at the same time. Eating a variety of food from all the different food groups in the proper amount and balance will supply the required nutrients that the body needs (PDRI, 2015).

The major nutrients that the body needs include:

- Macronutrients: proteins, carbohydrates, and fats
- Micronutrients: vitamins A, D, E, K, B complex; minerals such as calcium, iron, iodine, zinc, and fluoride
- Water and electrolytes

The following figures are derived from the Philippine Dietary Reference Intakes (2018):

Philippine Dietary Reference Intakes 2015: Summary Tables 1

Recommended Energy Intakes per day

Life stage/ age group	Weight (kg)		Energy (kcal)	
	M	F	M	F
Infants, mo				
0–5	6.5	6.0	620	560
6–11	9.0	8.0	720	630
Children, y				
1–2	12.0	11.5	1,000	920
3–5	17.5	17.0	1,350	1,260
6–9	23.0	22.5	1,600	1,470
10–12	33.0	36.0	2,060	1,980
13–15	48.5	46.0	2,700	2,170
16–18	59.0	51.5	3,010	2,280
Adults, y				
19–29	60.5	52.5	2,530	1,930
30–49	60.5	52.5	2,420	1,870
50–59	60.5	52.5	2,420	1,870
60–69	60.5	52.5	2,140	1,610
≥ 70	60.5	52.5	1,960	1,540
Pregnant				+300*
Lactating				+500

Abbreviations: M–Male, F–Female, E–Energy
*For 2nd and 3rd trimesters only

Acceptable Macronutrient Distribution Ranges

Lifestage/ age group	Range (% of Energy)		
	Protein	Total Fat	Carbohydrate*
Infants, mo			
0–5	5	40–60	35–55
6–11	8–15	30–40	45–62
Children, y			
1–2	6–15	25–35	50–69
3–18	6–15	15–30	55–79
Adults, y			
≥ 19	10–15	15–30	55–75

NOTE: *Acceptable Macronutrient Distribution Range (AMDR)* is the range of intakes for a particular energy source (carbohydrate, protein or fat) that is associated with reduced risk of chronic diseases while providing adequate intakes of essential nutrients, expressed as a percentage of total energy intake.
*The AMDR for carbohydrate is the percentage of total energy available after taking into account that consumed as protein and fat, hence the wide ranges.

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Figure 11. Daily recommended Energy Intake

Figure 6 shows the recommended daily intake of energies across different sex, weight, and age groups. The recommended energy is further divided into three types in the *acceptable macronutrient distribution ranges*. It is done to make a recommended intake of these types of sources of energy that reduces risks of chronic diseases.

Recommended Nutrient Intakes per day (Macronutrients)

Life stage/ age group	Weight (kg)		Energy (kcal)		Protein (g)		Essential Fatty Acids		Dietary Fiber (g)	Water (mL)	
							α-Linolenic Acid (%E)	Linoleic Acid (%E)			
	M	F	M	F	M	F				M	F
Infants, mo											
0–5	6.5	6.0	620	560	**9**	**8**	*0.5*	*4.5*	-	*680*	*680*
6–11	9.0	8.0	720	630	**17**	**15**	*0.5*	*4.5*	-	*890*	*890*
Children, y											
1–2	12.0	11.5	1,000	920	**18**	**17**	*0.5*	*3.0*	*6–7*	*1,000*	*920*
3–5	17.5	17.0	1,350	1,260	**22**	**21**	*0.5*	*2.0*	*8–10*	*1,350*	*1,260*
6–9	23.0	22.5	1,600	1,470	**30**	**29**	*0.5*	*2.0*	*11–14*	*1,600*	*1,470*
10–12	33.0	36.0	2,060	1,980	**43**	**46**	*0.5*	*2.0*	*15–17*	*2,060*	*1,980*
13–15	48.5	46.0	2,700	2,170	**62**	**57**	*0.5*	*2.0*	*18–20*	*2,700*	*2,170*
16–18	59.0	51.5	3,010	2,280	**72**	**61**	*0.5*	*2.0*	*21–23*	*3,010*	*2,280*
Adults, y											
19–29	60.5	52.5	2,530	1,930	**71**	**62**	*0.5*	*2.0*	*20–25*	*2,530*	*1,930*
30–49	60.5	52.5	2,420	1,870	**71**	**62**	*0.5*	*2.0*	*20–25*	*2,420*	*1,870*
50–59	60.5	52.5	2,420	1,870	**71**	**62**	*0.5*	*2.0*	*20–25*	*2,420*	*1,870*
60–69	60.5	52.5	2,140	1,610	**71**	**62**	*0.5*	*2.0*	*20–25*	*2,140*	*1,610*
≥ 70	60.5	52.5	1,960	1,540	**71**	**62**	*0.5*	*2.0*	*20–25*	*1,960*	*1,540*
Pregnant				+300*		**+27**					*+300*
Lactating				+500		**+27**					*+700*

NOTE: Recommended Nutrient Intakes (RNI) are in **bold font,** while Adequate Intakes (AI) are in *italics.*
*For 2nd and 3rd trimesters only

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Figure 12. Daily recommended Nutrient Intakes (Macronutrients)

Figure 7 shows the recommended daily intakes of macronutrients needed by the body across different sex, weight, and age groups. Energy, protein, fatty acids, dietary fiber, and water are displayed with their quantity in grams, kilograms, kilocalories, percent of Energy, and milliliters accordingly.

Recommended Nutrient Intakes per day (Vitamins)

Life stage/ age group	Weight (kg)		Vitamin A[a] (µgRE)		Vitamin D[b] (µg)		Vitamin E[c] (mg α-TE)		Vitamin K (µg)		Thiamin (mg)		Riboflavin (mg)		Niacin[d] (mgNE)		Vitamin B_6 (mg)		Vitamin B_{12} (µg)		Folate[e] (µgDFE)		Vitamin C (mg)	
	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F
Infants, mo																								
0–5	6.5	6.0	*380*	*380*	*5*	*5*	*3*	*3*	**7**	**6**	*0.2*	*0.2*	*0.3*	*0.3*	*1*	*1*	*0.1*	*0.1*	*0.3*	*0.3*	*65*	*65*	*30*	*30*
6–11	9.0	8.0	**400**	**400**	*5*	*5*	*4*	*4*	**9**	**8**	**0.4**	**0.3**	**0.4**	**0.3**	**5**	**5**	*0.2*	*0.3*	*0.4*	*0.4*	*80*	*70*	*40*	*40*
Children, y																								
1–2	12.0	11.5	**400**	**400**	*5*	*5*	*4*	*4*	**12**	**12**	**0.5**	**0.4**	**0.5**	**0.4**	**6**	**6**	**0.5**	**0.5**	**0.9**	**1.0**	**150**	**150**	**45**	**45**
3–5	17.5	17.0	**400**	**400**	*5*	*5*	*5*	*5*	**18**	**17**	**0.5**	**0.5**	**0.6**	**0.5**	**7**	**7**	**0.6**	**0.7**	**1.1**	**1.2**	**200**	**200**	**45**	**45**
6–9	23.0	22.5	**400**	**400**	*5*	*5*	*6*	*6*	**23**	**23**	**0.7**	**0.7**	**0.7**	**0.7**	**9**	**9**	**0.7**	**0.8**	**1.3**	**1.5**	**300**	**300**	**45**	**45**
10–12	33.0	36.0	**500**	**500**	*5*	*5*	*7*	*9*	**33**	**36**	**0.9**	**0.9**	**1.0**	**0.9**	**11**	**12**	**1.0**	**1.1**	**1.8**	**2.1**	**300**	**300**	**45**	**45**
13–15	48.5	46.0	**700**	**500**	*5*	*5*	*10*	*9*	**49**	**46**	**1.2**	**1.0**	**1.3**	**1.0**	**15**	**13**	**1.3**	**1.2**	**2.3**	**2.2**	**400**	**400**	**60**	**55**
16–18	59.0	51.5	**800**	**600**	*5*	*5*	*11*	*10*	**59**	**52**	**1.4**	**1.1**	**1.5**	**1.1**	**18**	**14**	**1.5**	**1.3**	**2.7**	**2.4**	**400**	**400**	**70**	**60**
Adults, y																								
19–29	60.5	52.5	**700**	**600**	*5*	*5*	*10*	*10*	**61**	**53**	**1.2**	**1.1**	**1.3**	**1.1**	**16**	**14**	**1.3**	**1.3**	**2.4**	**2.4**	**400**	**400**	**70**	**60**
30–49	60.5	52.5	**700**	**600**	*5*	*5*	*10*	*10*	**61**	**53**	**1.2**	**1.1**	**1.3**	**1.1**	**16**	**14**	**1.3**	**1.3**	**2.4**	**2.4**	**400**	**400**	**70**	**60**
50–59	60.5	52.5	**700**	**600**	*10*	*10*	*10*	*10*	**61**	**53**	**1.2**	**1.1**	**1.3**	**1.1**	**16**	**14**	**1.7**	**1.6**	**2.4**	**2.4**	**400**	**400**	**70**	**60**
60–69	60.5	52.5	**700**	**600**	*15*	*15*	*10*	*10*	**61**	**53**	**1.2**	**1.1**	**1.3**	**1.1**	**16**	**14**	**1.7**	**1.6**	**2.4**	**2.4**	**400**	**400**	**70**	**60**
≥ 70	60.5	52.5	**700**	**600**	*15*	*15*	*10*	*10*	**61**	**53**	**1.2**	**1.1**	**1.3**	**1.1**	**16**	**14**	**1.7**	**1.6**	**2.4**	**2.4**	**400**	**400**	**70**	**60**
Pregnant				**+300**		*+0*		*+0*		**+0**		**+0.3**		**+0.7**		**+4**		**+0.6**		**+0.2**		**+200**		**+10**
Lactating				**+400**		*+0*		*+4*		**+0**		**+0.2**		**+0.6**		**+3**		**+0.7**		**+0.5**		**+150**		**+35**

NOTE: Recommended Nutrient Intakes (RNI) are in **bold font,** while Adequate Intakes (AI) are in *italics*.

[a] 1 retinol equivalent (RE) = 1 µg retinol = 12 µg β-carotene or 24 µg other provitamin A carotenoids; 1 µg RE = 3.33 IU vitamin A

[b] In the absence of adequate exposure to sunlight, as calciferol;1 µg calciferol = 40 IU vitamin D

[c] 1 mg alpha-tocopherol equivalent (α-TE) = 1.49 IU natural form or 2.22 IU synthetic form

[d] As niacin equivalent (NE)

[e] 1 dietary folate equivalent (DFE) = 1 µg food folate = 0.6 µg folic acid from fortified foods or as supplement = 0.5 µg taken on an empty stomach

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Figure 13. Daily recommended Nutrient Intakes (Vitamins)

Figure 8 shows the recommended daily intakes of vitamins needed by the body across different sex, weight, and age groups. Primary essential vitamins that our bodies need is displayed with their quantity in grams, micrograms, and milligrams accordingly.

Recommended Nutrient Intakes per day (Minerals)

Life stage/ age group	Weight (kg)		Iron (mg)		Zinc (mg)		Selenium (µg)		Iodine (µg)		Calcium (mg)		Magnesium (mg)		Phosphorus (mg)		Fluoride (mg)		Electrolytes		
																			Sodium	Chloride	Potassium
	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	M	F	(mg)	(mg)	(mg)
Infants, mo																					
0–5	6.5	6.0	*0.4*	*0.4*	*2.0*	*2.0*	**7**	**6**	**90**	**90**	*200*	*200*	*26*	*26*	*90*	*90*	*0.01*	*0.01*	*120*	*180*	*500*
6–11	9.0	8.0	**10**	**9**	**4.2**	**3.7**	**10**	**9**	**90**	**90**	**400**	**400**	*50*	*50*	*275*	*275*	*0.5*	*0.4*	*200*	*300*	*700*
Children, y																					
1–2	12.0	11.5	**8**	**8**	**4.1**	**4.0**	**17**	**16**	**90**	**90**	**500**	**500**	**60**	**60**	**460**	**460**	*0.6*	*0.6*	*225*	*350*	*1,000*
3–5	17.5	17.0	**9**	**9**	**5.0**	**4.8**	**20**	**20**	**90**	**90**	**550**	**550**	**70**	**70**	**500**	**500**	*0.9*	*0.9*	*300*	*500*	*1,400*
6–9	23.0	22.5	**10**	**9**	**5.1**	**5.0**	**20**	**19**	**120**	**120**	**700**	**700**	**90**	**90**	**500**	**500**	*1.2*	*1.1*	*400*	*600*	*1,600*
10–12	33.0	36.0	**12**	**20**	**6.6**	**6.1**	**21**	**23**	**120**	**120**	**1,000**	**1,000**	**150**	**160**	**1,250**	**1,250**	*1.7*	*1.8*	*500*	*750*	*2,000*
13–15	48.5	46.0	**19**	**(28)**	**9.2**	**7.4**	**30**	**29**	**150**	**150**	**1,000**	**1,000**	**220**	**210**	**1,250**	**1,250**	*2.4*	*2.3*	*500*	*750*	*2,000*
16–18	59.0	51.5	**14**	**(28)**	**9.0**	**7.2**	**37**	**32**	**150**	**150**	**1,000**	**1,000**	**265**	**230**	**1,250**	**1,250**	*3.0*	*2.6*	*500*	*750*	*2,000*
Adults, y																					
19–29	60.5	52.5	**12**	**(28)**	**6.5**	**4.6**	**38**	**33**	**150**	**150**	**750**	**750**	**240**	**210**	**700**	**700**	*3.0*	*2.6*	*500*	*750*	*2,000*
30–49	60.5	52.5	**12**	**(28)**	**6.5**	**4.6**	**38**	**33**	**150**	**150**	**750**	**750**	**240**	**210**	**700**	**700**	*3.0*	*2.6*	*500*	*750*	*2,000*
50–59	60.5	52.5	**12**	**10**	**6.5**	**4.6**	**38**	**33**	**150**	**150**	**750**	**800**	**240**	**210**	**700**	**700**	*3.0*	*2.6*	*500*	*750*	*2,000*
60–69	60.5	52.5	**12**	**10**	**6.5**	**4.6**	**38**	**33**	**150**	**150**	**800**	**800**	**240**	**210**	**700**	**700**	*3.0*	*2.6*	*500*	*750*	*2,000*
≥ 70	60.5	52.5	**12**	**10**	**6.5**	**4.6**	**38**	**33**	**150**	**150**	**800**	**800**	**240**	**210**	**700**	**700**	*3.0*	*2.6*	*500*	*750*	*2,000*
Pregnant				**(+10)**		**+5.1**		**+4**		**+100**		**+50***		**+0**		**+0**		*+0*	-	-	-
Lactating				**+2**		**+7.0**		**+9**		**+100**		**+0**		**+50**		**+0**		*+0*	-	-	-

NOTE: Recommended Nutrient Intakes (RNI) are in **bold font,** while Adequate Intakes (AI) are in *italics.*

( ) Requirements cannot be met by usual diet alone. Intake of iron-rich and iron-fortified foods and the use of supplements are recommended, if necessary.

*The calcium recommendation for pregnant women is for 3rd trimester only.

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Figure 14. Daily recommended Nutrient Intakes (Minerals)

Figure 9 shows the recommended daily intakes of minerals needed by the body across different sex, weight, and age groups. Primary essential minerals that our bodies need is displayed with their quantity in micrograms, and milligrams accordingly.

2.8.2 Food Pyramid

The food pyramid was made to better understand the nutritional guidelines. By translating the guidelines to a graphic presentation.

Usually, Filipino diet consists mainly of rice. It contributes a large part of the carbohydrate intake of Filipinos. Thus, carbohydrate foods are placed at the base of the pyramid and are for liberal consumption to meet the energy needs. Vegetables and fruits comprise the second level of the pyramid. The third level consists of animal protein foods such as fishes, meats, poultries, seafoods, milk, dairies, dried beans, and nuts. These foods are recommended to be consumed moderately for these foods provide high quality

proteins, vitamins, and minerals. Lastly, at the tip of the pyramid, the fats and oils reside that are recommended to be taken just enough. Water is also present at the bottom of the pyramid as it is an essential nutrient that the body needs.

Figure 15. Daily Nutritional Guide Pyramid for Filipino Children (1-6 years old)

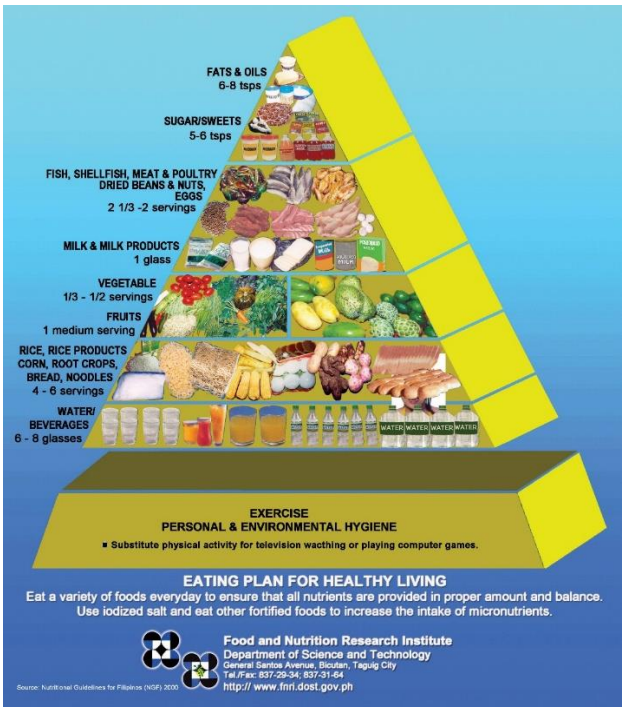


Figure 16. Daily Nutritional Guide Pyramid for Filipino Children (7-12 years old)

Figure 17. Daily Nutritional Guide Pyramid for Filipino Teenager (13-19 years old)

Figure 18. Daily Nutritional Guide Pyramid for Filipino Adults (20-39 years old)

Figure 19. Daily Nutritional Guide Pyramid for Filipino Adults (60-69 years old)

Figures 10-14 depicts the graphic-translated nutritional guides for Filipinos. To make food intake recommendations easier to understand.

2.8.3 Pinggang Pinoy (Healthy food plates for Filipinos)

The Pinggang Pinoy is another project by the Food and Nutrition Research Institute of the Department of Science and Technology of the Philippines that simplifies both the nutritional guidelines and the nutritional food pyramid. The pinggang pinoy utilizes the Go, Grow, Glow method, where foods are classified within these three food groups.

According to an article (“Go, Grow, Glow Foods,” 2010), “Go” foods are foods that are rich in carbohydrate such as rice, sweet potatoes, pastas, and breads. This type of food provides the energy needed by a child. “Grow” foods are protein-rich foods that helps build bones and teeth, strengthens muscles, and improves mental functions such as meat, eggs, fish, and dairy products. “Glow” foods are foods that are filled with vitamins and

minerals that strengthens the immune system. Glow foods includes all kinds of fruits such as apples, oranges, bananas, avocados, etc.

Classification of foods into Go, Grow, Glow, is also supported by About Filipino Food ("Pinggang Pinoy", 2022) by defining Go foods with an example of rice, Glow foods includes both fruits and vegetables, and Grow foods with fishes.

Both these articles have their sources taken from the FNRI itself.

2.9 The Filipino Diet

Angeles-Agdeppa et al. (2019) said that Filipino diet suffers from the issue of limited diversity where breads, white rice, and pork contribute to most of the daily intake of nutrients like protein, energy, carbohydrates, and many more. Nutrient-packed food groups such as vegetables, fruit, and dairy are lacking in diet. Though they are the top two sources of vitamin C and folate in foods, less than 70% of the population daily ate vegetables and fewer than 25% ate fruits. This amount of consumption was not enough to support the adult's nutrition needs.

Also, another study by Angeles-Agdeppa and Custodio (2020) shows that the top-5 most consumed food are *Refined rice*, *pork*, *fish*, *fats* and *oil*, and other *vegetables*. Following that are the top 6-10 namely *chickens*, *egg and egg dishes*, other *sweet beverages*, *dark green leafy* and *deep-yellow vegetables*. The five most consumed food as a source of energy are rice, pork, fats and oils (mostly those that came from plants), chicken, and bread. A lot of carbohydrates (50%) came solely from *rice,* then *bread*, *sweetened beverages*, *noodles*, and then *soft drinks*. The top five protein contributors include *Refined rice*, *pork*, *fish*, *chicken*, *egg* and *egg dishes*. While 40% of the total fat intake from food intakes came from *pork*, the other sources were *fats and oils (from plant sources)*, *chicken*,

sausages, and *egg*. Close to half (42%) of the total *vitamin C* is from fruits and dark-green leafy vegetables but still there is still high prevalence of inadequate levels of vitamin C.

2.10 Synthesis

Convolutional Neural Network is an algorithm that takes image as an input together with weighted values and biases that help to differentiate other images. However, CNN consumes a lot of resources the more process it must go through. The proponents decided to use MobileNetV3 as the CNN model due to its smaller size and is tuned for mobile use which is perfect for the Foodietection application. CNN in general is very popular in image recognition as it can detect and recognize images with great accuracy. However, it is hard for CNN to classify an object without a distinct feature. As defined earlier in Pinggang Pinoy, Go, Grow, Glow are foods that are classified through their nutritional content and not their physical appearance. Taking for example the Glow food category, it consists of fruits and vegetables like Mango, apple, banana, squash, cabbages, carrots, and many more. All of them have physical features that are different from each other, which makes it impossible for CNN to classify them into food groups. Therefore, the proponents proposed a method that can classify food that is based on their nutrient contents. To fulfill this gap in food group classification, researchers will utilize the use of fuzzy logic. The use of Fuzzy Logic is a helpful approach that uses “Degrees of Truth” as opposed to the usual True or False (1 for true, 0 for false) Boolean logic together with the use of Fuzzy Rules like IF THEN rules. It will be used in the study to determine whether a food is applicable as Go, Grow, and Glow based on the degree of truth analyzed through the fuzzy rules that the proponents have set. The basis for the rules will be made through the help of a Nutritional Guide by Philippine Dietary Reference Intakes (2015) and other sources of nutritional guides.

Currently, there are no existing studies that exhaustively defines nutritional characteristics for the food groups Go, Grow, and Glow. Thus, it is imperative that the researchers find primary sources of information directly from food nutritionists, or any other experts regarding the subject matter. Our proposed system aims to introduce definitive nutritional characteristics for Go, Grow, and Glow foods.

Chapter 3

METHODOLOGY

In this chapter, the proponents presented the methodologies and techniques used in the implementation of this study. This chapter also shows the different processes used to obtain and organize data through different methods. How the proponents treated data will also be shown in the statistical treatments.

3.1 Research Design

The researchers will use quantitative design, specifically experimental design, for the study. Experimental design is a systematic and scientific approach to research in which the researcher manipulates one or more variables, and controls and measures any change in other variables (Blakstad, 2008). Experimental design method allows researchers to better understand and evaluate the factors that influence a particular system by means of statistical approaches (Hanrahan, 2005). The researchers believe that experimental design is the best for the study since the researchers will want to determine the accuracy of the system that will be developed.

3.2 Sources of Data

To thoroughly evaluate the proposed approach, we chose five types of datasets: (1) Food101 (Bossard et al., 2014): a dataset commonly used for benchmarking food classification accuracy. Purposefully, the images included in this dataset is not selected and checked by a human. The dataset contains 75,750 images with sometimes wrong labels and intense colors. The dataset has a total of 101,000 real-world images, including very diverse but also visually and semantically similar food classes. This allows us to validate our approach on a large dataset build with weakly labeled data (2) Our second dataset will be a dataset consisting of Filipino food (Hatfield, 2020) scraped from google images (Hatfield S., 2020); (3) Our third dataset is Fruits 360. It contains 90483 images of

101 types of foods and vegetables. Data distribution for this dataset will be 75% for training and 25% for testing. (4) Finally, the system will use NutriVision, which utilizes Nutritionix, a large database which has 700K food items that range from grocery foods, common foods, and restaurant dishes for nutritional values that the Philippine FNRI does not have.

3.2.1 Data Composition

The proponents have four (4) data sets namely *Food101*, *FilipinoFoods* (from Google images), *FilipinoFoods* (from Google images by the Researchers), and *Fruits360* for training the images. The datasets will be divided into 80% Training data (including 10% Validation data) and Testing data of 20%.

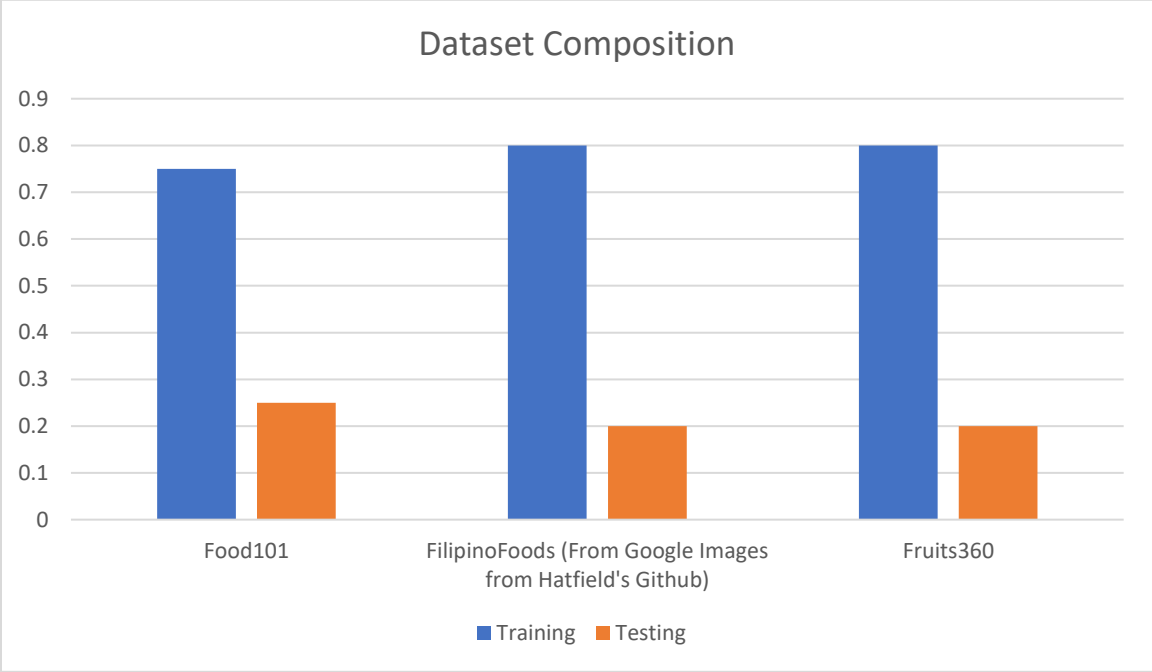


Figure 20. Composition of the Datasets

3.3 Research Instruments

Interview – The researchers will use interview to gather correct nutritional values for go, grow, and glow foods from licensed nutritionists (from the FNRI) and dietitians from health centers around Metro Manila.

Experiment Paper – This is where the researchers gather the results of the testing of the images, the food group classifier, and the recommendation system of the application. After testing for the accuracy of the mobilenetv3 and the fuzzy logic system that the researchers will create, the researchers will then apply the same approach to the mobilenetv2 to gather information regarding the significant difference of the two models.

Foodietection Application – this will be the application to be developed for the users. This include features such as capturing images using the camera of the android phone of the user, providing nutritional information of the scanned foods, Go, Grow, Glow indicator that shows if the food is a Go, Grow, Glow and if the meal has the complete set of Go, Grow, Glow. Also, the app provides a recommended food intake to fill the missing nutrients of the current captured image of the application based on the FNRI nutritional guide.

3.3.1 Development Details

Android Studio – The official integrated development environment for Google's Android operating system, built on JetBrains' IntelliJ IDEA software and designed specifically for Android development. The researchers will use Android Studio as their tool emulator.

React Native – React Native is an open-source UI software framework created by Meta Platforms, Inc. It is used to develop applications for Android, Android TV, iOS, macOS, tvOS, Web, Windows and UWP by enabling developers to use the React framework along with native platform capabilities. The researchers will use React Native as their primary language for GUI development.

Visual Studio Code – Visual Studio Code, also commonly referred to as VS Code, is a source-code editor made by Microsoft with the Electron Framework, for Windows,

Linux and macOS. Features include support for debugging, syntax highlighting, intelligent code completion, snippets, code refactoring, and embedded Git. The researchers will use VS Code as their main IDE for coding the tool.

TensorFlow – is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks. The researchers will use TensorFlow to access machine learning tools that will be used for the application.

Keras – is an open-source software library that provides a Python interface for artificial neural networks. Keras acts as an interface for the TensorFlow library.

3.3.2 Developmental Process

The researchers are planning to use Visual Studio Code as their main IDE to develop the system. Then Android Studio and React Native will be used for the android application. The researchers are also planning to use Keras for image processing. Finally, IBM's Statistical Package for the Social Sciences (SPSS) is going to be used for the statistical treatment of data.

Software Development Life Cycle (SDLC)

The proponents of this study plan to use the Waterfall Approach or Waterfall Model for the system development. Waterfall Model has 5 principal phases that covers the fundamental product activities. This part showed what was planned for each development phase.

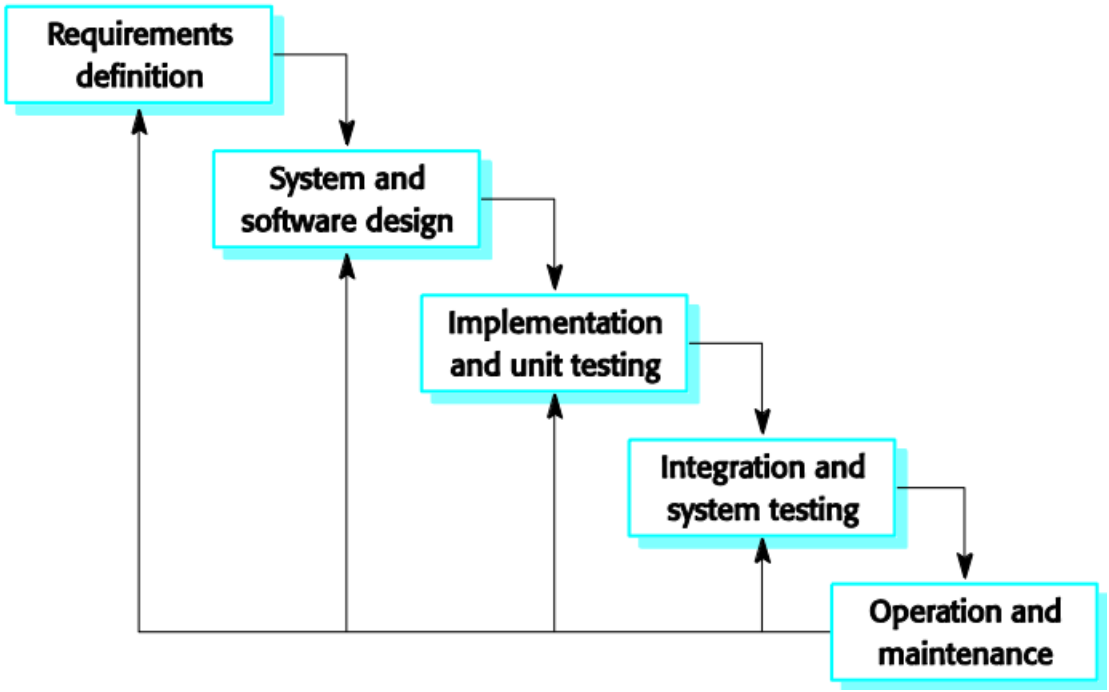


Figure 21. The Waterfall Model

1) *Analysis* – The proponents first gathered information about the relevant literatures and studies for Food Recognition and Go, Grow, Glow Foods. After establishing the related literature and studies, the proponents looked through multiple food datasets to be used for the system. Next is to determine the software that the proponents will use in developing the system. Then finally determining the way to rate the system according to its Accuracy, Precision, and F-1 Score.
2) *System/Software Design* – In the second phase, the proponents proposed their System Architecture (see Figure 18) and the proposed interface design. The proposed system design will consist of a Navigation bar where all the reachable pages are shown such as the Home, Capture, Search and Profile pages and all their consecutive pages (see Appendix B for full image details of the prototype).
3) *Implementation* – In this phase, the proponents will create the working mobile application with the use of Android Studio and React Native. Then use python to

be able to use Keras library to use MobileNetv3 for image detection. Then images from the datasets are going to be used to train the image detection model.

4) *Testing and Integration* – After creating the initial mobile application, the proponents will test the created system using the different datasets. Initial testing will be done to determine the effectiveness of the proposed system.
5) *Operation and Maintenance* – In the final phase, the final developed mobile application will be tested to determine whether the proposed system is working as intended.

3.4 Data Gathering Procedure

The researchers will start with building the image object detection through experimenting with different batch sizes and steps for the current architecture. This specific procedure will get data of the optimal architecture we can use given the limitation of the training devices to be used. The researchers also took previously used datasets of computer vision-related studies. The researchers also utilize cloud-based databases for food information, specifically the Nutritionix database. Using the Nutritionix DATABSE, the researchers will first, find patterns in the nutritional values of several foods included in each food group of the Go, Grow, Glow foods. After finding the patterns that distinguish Go, Grow, Glow foods have in their nutritional values. We will then use this pattern as a ruleset for the fuzzy logic of our recommendation system. We will refer to food nutritionists to check for the accuracy of our classifier of food into go, grow, glow. Before finally using the fuzzy logic again for the recommendation system with the FNRI food intake guidelines as a ruleset for the approximated daily food intakes of users with respect to their age, weight, and sex.

3.5 Ethical Consideration

This study aims to help individuals in achieving the recommended daily food intakes as per the recommendations made by the Philippine Dietary Reference Intakes. Every information from this study's application will be an approximation given by the user's information. Every recommendation is based on the user's information such as age, body weight, and sex and the PDRI's nutritional guide. Thus, the application will not be used to discriminate users by their demographics such as their age or weight but will instead use it properly to give an accurate approximation of what they should be taking as an aid to have a proper diet.

3.6 Proposed System Architecture

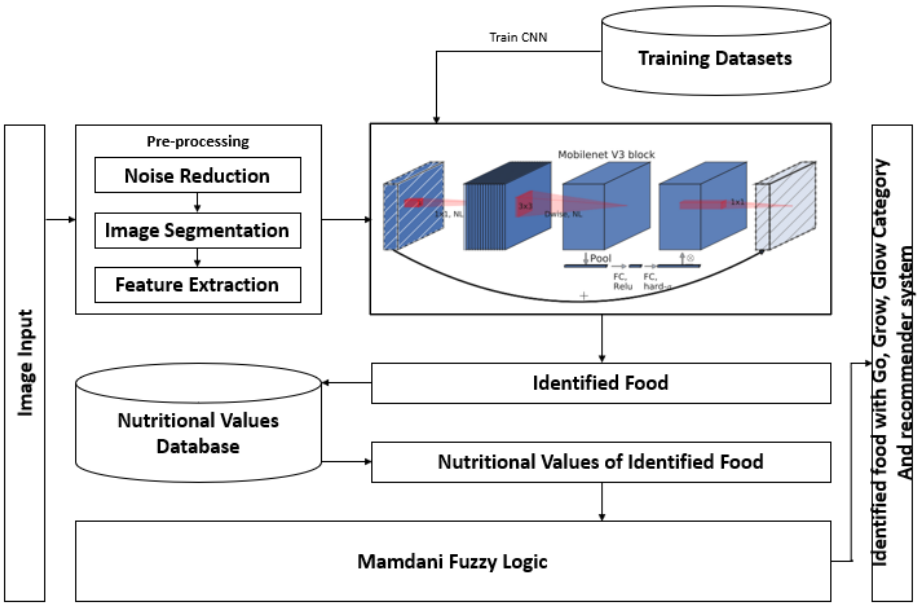


Figure 22. System Architecture

Figure 22 shows the system architecture that will be used.

3.7 Statistical Treatment of Data

To test the proposed system of the proponents, the following statistical formulas were used:

	ACTUAL CLASS TRUE (1) OR FALSE (0)	
TRUE (1) PREDICTED CLASS FALSE (0)	TP (1, 1)	FP (1, 0)
	FN (0, 1)	TN (0, 0)

Figure 23. Confusion Matrix

Where:

TP – True Positive

FP – False Positive

FN – False Negative

TN – True Negative

Confusion Matrices are used to visualize analytics like *Precision*, *Recall*, and *Accuracy*. They are useful because they give the direct values of comparison like True Positives, True Negatives, False Positives, and False Negatives

Precision is used to get the percentage of "If the system predicts true, how often is it correct?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

Formula 1. Precision Formula

$$\text{Recall} = \frac{TP}{TP + FN}$$

Formula 2. Recall Formula

Recall, also known as **True Positive Rate,** is used to get the value of “When the input is actually true, how often does the system predict true?”

$$F_1 = 2 * \frac{precision * recall}{precision + recall}$$

Formula 3. F1 Score Formula

F1 Score is the harmonic average of Precision and Recall and F1 score takes both FP and FN into account.

$$Accuracy = \frac{\#\ of\ correct\ predictions}{\#\ of\ total\ predictions}$$

Formula 4. Accuracy Formula

Accuracy formula is used to check if the proposed system can detect and classify Go, Grow, Glow foods correctly. To get the percentage, multiply the result by 100. To check, a table with verbal interpretation of accuracy is used.

Rating	Verbal Interpretation of Accuracy
75.01-100%	Very High Accuracy
50.01-75.00%	High Accuracy
25.01-50.00%	Low Accuracy
0-25.00%	Very Low Accuracy

Figure 24. Verbal Interpretation of Accuracy (Zawbaa, Abbass, Basha, Hazman, & Hassenian, 2014)

Two-sample t-test (also known as the independent samples *t*-test) is a method used to test whether the unknown population means of two groups are equal or not.

$$t = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Formula 5. Two-Sample t-test

APPENDICES

Appendix A: Experiment Paper

Experiment Paper #1

No. of Test	Evaluation of System's Performance				Precision %	Recall %	F1 Score %	Verbal Interpretation
	TP	FP	TN	FN				

Experiment Paper #2

Image Input	Trial					No. Correct Classification	Accuracy %
	1	2	3	4	5		
		/					

Experiment Paper #3

Image Input	Food Group (Classification Prediction)					No. Correct Classification	Accuracy %
	1	2	3	4	5		
Go							

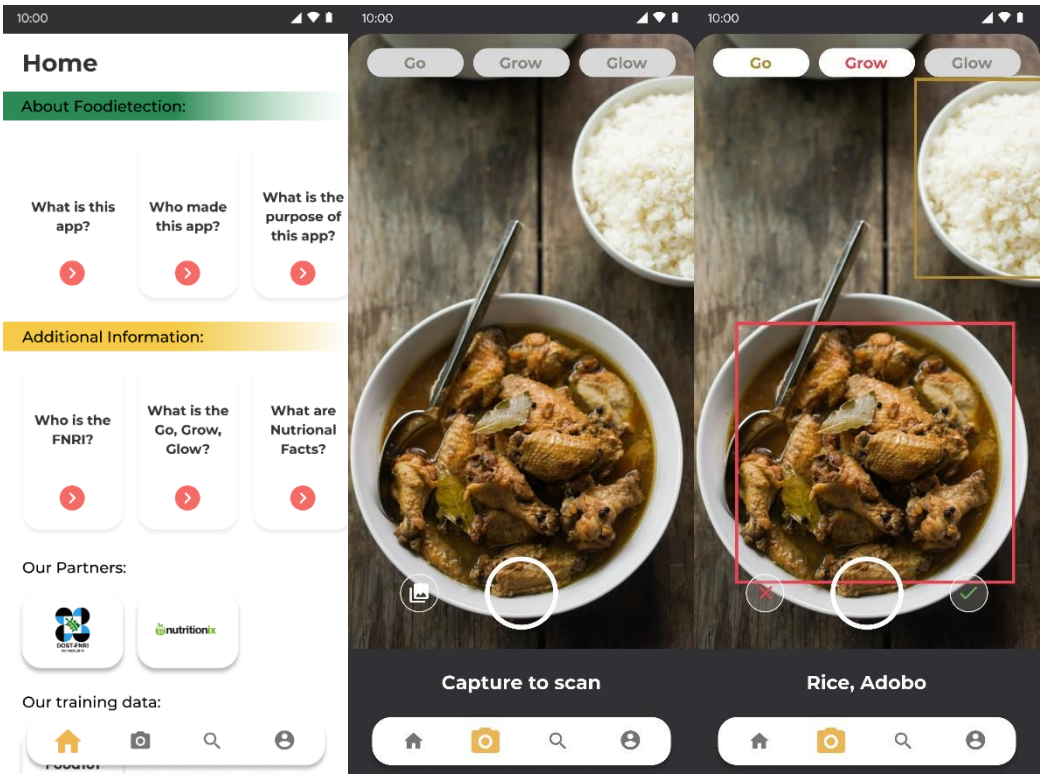
Image Input	Food Group (Classification Prediction)					No. Correct Classification	Accuracy %
	1	2	3	4	5		
Grow							

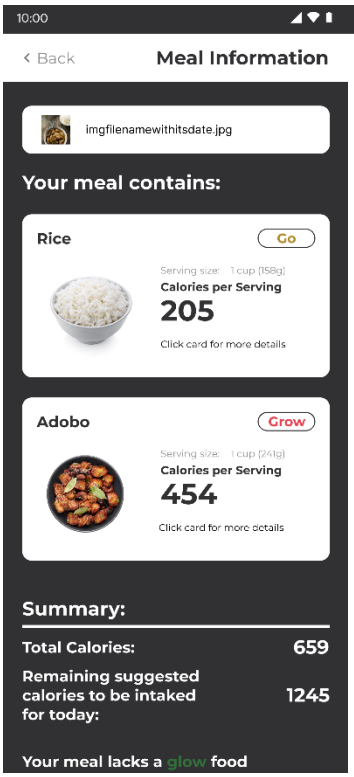
Image Input	Food Group (Classification Prediction)					No. Correct Classification	Accuracy %
	1	2	3	4	5		
Glow							

Experiment Paper #4

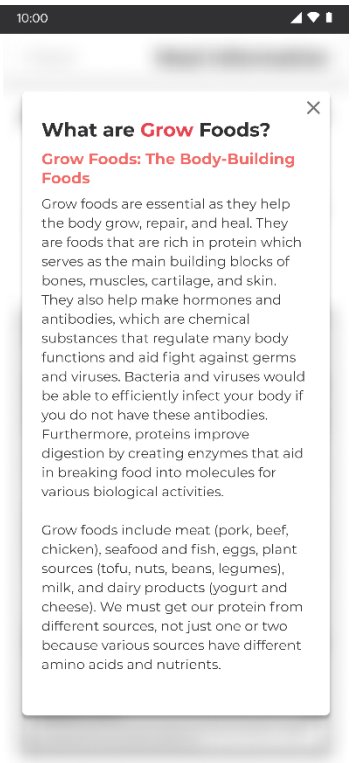
Image Input	Trial					No. Correct Classification	Accuracy %
	1	2	3	4	5		

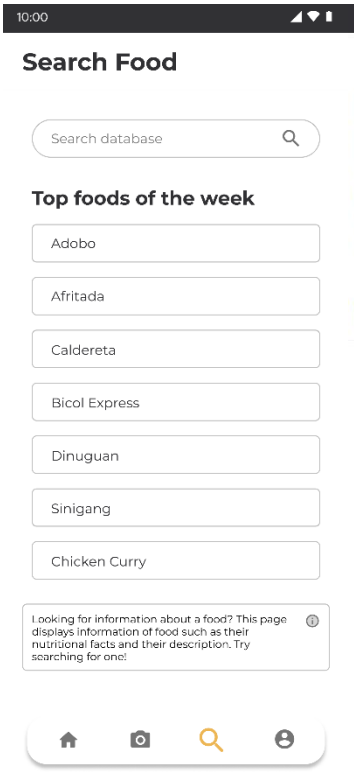
Appendix B: System Design

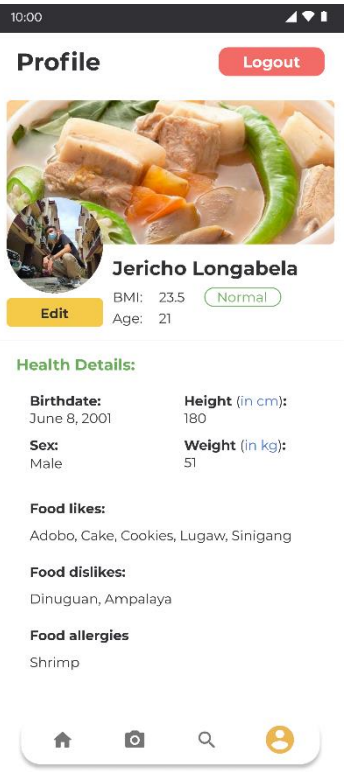



10:00
Back
Meal Information
imgfilenamewithitsdate.jpg
Your meal contains:
Rice
Go
Serving size: 1 cup (158g)
Calories per Serving
205
Click card for more details
Adobo
Grow
Serving size: 1 cup (241g)
Calories per Serving
454
Click card for more details
Summary:
Total Calories: 659
Remaining suggested calories to be intaked for today: 1245
Your meal lacks a glow food

10:00
Back
Meal Information
Adobo
Grow
Serving size: 1 cup (241g)
Calories per Serving
454
Nutrition Facts
2 Servings per container
Serving size 1 cup (140g)
Amount per serving
Calories 160
% Daily Value*
Total Fat 8g 10%
Saturated Fat 8g 10%
Trans Fat 8g
Cholesterol 0mg 0%
Sodium 60mg 3%
Total Carbohydrate 21g 8%
Dietary Fiber 3g 11%
Total Sugars 15g
Includes 5g added sugars 10%
Protein 3g 10%
Vitamin D 5mcg 25%
Calcium 20mg 2%
Iron 1mg 6%
Potassium 230mg 4%
*The % Daily Value tells you how much a nutrient in a serving food contributes to a daily die. 2000 calories a day is used for general nutrition advice.


10:00
What are Grow Foods?
Grow Foods: The Body-Building Foods
Grow foods are essential as they help the body grow, repair, and heal. They are foods that are rich in protein which serves as the main building blocks of bones, muscles, cartilage, and skin. They also help make hormones and antibodies, which are chemical substances that regulate many body functions and aid fight against germs and viruses. Bacteria and viruses would be able to efficiently infect your body if you do not have these antibodies. Furthermore, proteins improve digestion by creating enzymes that aid in breaking food into molecules for various biological activities.
Grow foods include meat (pork, beef, chicken), seafood and fish, eggs, plant sources (tofu, nuts, beans, legumes), milk, and dairy products (yogurt and cheese). We must get our protein from different sources, not just one or two because various sources have different amino acids and nutrients.


10:00
Search Food
Search database
Top foods of the week
Adobo
Afritada
Caldereta
Bicol Express
Dinuguan
Sinigang
Chicken Curry
Looking for information about a food? This page displays information of food such as their nutritional facts and their description. Try searching for one!


10:00
Profile
Logout
Jericho Longabela
Edit
BMI: 23.5
Normal
Age: 21
Health Details:
Birthdate:
June 8, 2001
Height (in cm):
180
Sex:
Male
Weight (in kg):
51
Food likes:
Adobo, Cake, Cookies, Lugaw, Sinigang
Food dislikes:
Dinuguan, Ampalaya
Food allergies
Shrimp

Appendix C: List of Food Types

Food Type	Corresponding Dataset
1) Apple	Fruits360
2) Arroz Caldo	FilipinoFoods
3) Avocado	Fruits360
4) Balut	FilipinoFoods
5) Banana	Fruits360
6) Bicol Express	FilipinoFoods
7) Bulalo	FilipinoFoods
8) Champorado	FilipinoFoods
9) Cherry	Fruits360
10) Chicharon	FilipinoFoods
11) Chicken Adobo	FilipinoFoods
12) Chicken Wings	Food101
13) Crispy Pata	FilipinoFoods
14) Fried Rice	Food101
15) Grapes	Fruits360
16) Halo-halo	FilipinoFoods
17) Kaldereta	FilipinoFoods
18) Kiwi	Fruits360
19) Laing	FilipinoFoods
20) Leche Flan	FilipinoFoods
21) Lechon	FilipinoFoods
22) Lemon	Fruits360
23) Liempo	FilipinoFoods
24) Longganisa	FilipinoFoods
25) Lumpia	FilipinoFoods
26) Mango	Fruits360
27) Orange	Fruits360
28) Pancit	FilipinoFoods
29) Pandesal	FilipinoFoods
30) Papaya	Fruits360
31) Pear	Fruits360
32) Pinakbet	FilipinoFoods
33) Pineapple	Fruits360
34) Pork Adobo	FilipinoFoods
35) Pork Afritada	FilipinoFoods
36) Rambutan	Fruits360
37) Sisig	FilipinoFoods
38) Tinolang Manok	FilipinoFoods
39) Turon	FilipinoFoods

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